

Re-investigate *Word Concreteness Effect* Using Z-Curve Analysis

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Introduction

Publication Bias & Z-Curve

A basic problem: A published literature shaped by selection cannot be trusted at face value
— Z-curve allows us to ask how credible and replicable the published evidence is

The Credibility Problem

- ▶ Questionable Research Practices (QRPs) distort reported findings: selective stopping, cherry-picking conditions, covariate fishing
- ▶ Publication bias favors $p < .05$; null and ambiguous results are suppressed
- ▶ A nominal 5% false-positive rate can inflate to 61% under combined QRPs (Simmons, Nelson & Simonsohn, 2011)
- ▶ Consequence: published effects are inflated and poorly replicable (36% replication rate in OSC, 2015)

Why Z-Curve, Not Just P-Curve

Models the latent distribution of statistical power in a literature **using only significant z-values**

Three indicators:

- ▶ **EDR (Expected Discovery Rate)**: estimated share of all conducted studies (published + unpublished) reflecting true effects
- ▶ **ERR (Expected Replication Rate)**: average power of the published significant findings
- ▶ **EFDR (Expected False Discovery Rate)**: proportion of published significant findings that are likely false positives

Pre-registration Summary

AsPredicted Registration

- Research Question
 - To what extent does the published literature on the concreteness effect in word recognition reflect reliable and replicable findings, as assessed through Z-curve analysis of observed and expected discovery rates?
- Dependent Variables
 - ODR, EDR, ERR, EFDR
- Analysis: Extract p-values or test statistics → convert to absolute z-scores → fit z-curve model via the *Zcurve* R package
- Inclusion criteria
 - Empirical studies & quantitative behavioral data studying the word recognition task using healthy human participants, with sufficient statistics for effect size calculation
 - Published after 2000→ revised to 1980 due to insufficient number of papers
- Sample size: ~50 papers

Methods

Word Concreteness Effect

The concreteness effect refers to the well-documented phenomenon where **concrete words** (e.g., **table, mountain**) are recognized faster than **abstract words** (e.g., justice, theory) (Paivio & Begg, 1971; Wickens & Engle, 1970). Studies within this domain typically employ behavioral measures such as reaction times and accuracy rates to examine how concreteness of words influence the speed and accuracy of word recognition. This line of inquiry contributes to our broader understanding of the mental lexicon, semantic memory organization, and the embodied nature of language processing .

Literature Selection Criteria

- **Study design:** Empirical research with quantitative data (e.g., reaction times, accuracy rates, error rates).
- **Participants:** Participants are human adults who are healthy and free from reported neurological, psychiatric, or language-related diagnoses.
- **Manipulation of concreteness:** The study directly manipulates or compares word concreteness, either as a categorical variable (concrete vs. abstract words) or as a continuous variable.
- **Task:** The study employs tasks that require word recognition, such as lexical decision, word naming, semantic categorization, or word identification.
- **Statistical reporting:** The study reports sufficient statistical information to extract or calculate effect sizes (e.g., means with SDs, t-values with df, F-values with df, or direct effect size estimates).
- **Publication year:** The study was published in or after 1980.

Focal Hypothesis Tests

- EDR of published findings is significantly lower than ODR
- ERR is significantly lower than ODR
- EFDR substantially exceeds the nominal $\alpha = 5\%$

Data Extraction Procedures

- Two primary outcomes extracted: reaction times (ms) and accuracy (proportion correct)
- Coded for each study: title, author, year, DOI, study design, sample size, test type, test statistic, df , p-value, and effect direction
- Test statistics (t , F , r , χ^2 , z) converted to z-scores for z-curve analysis

How to treat p-values

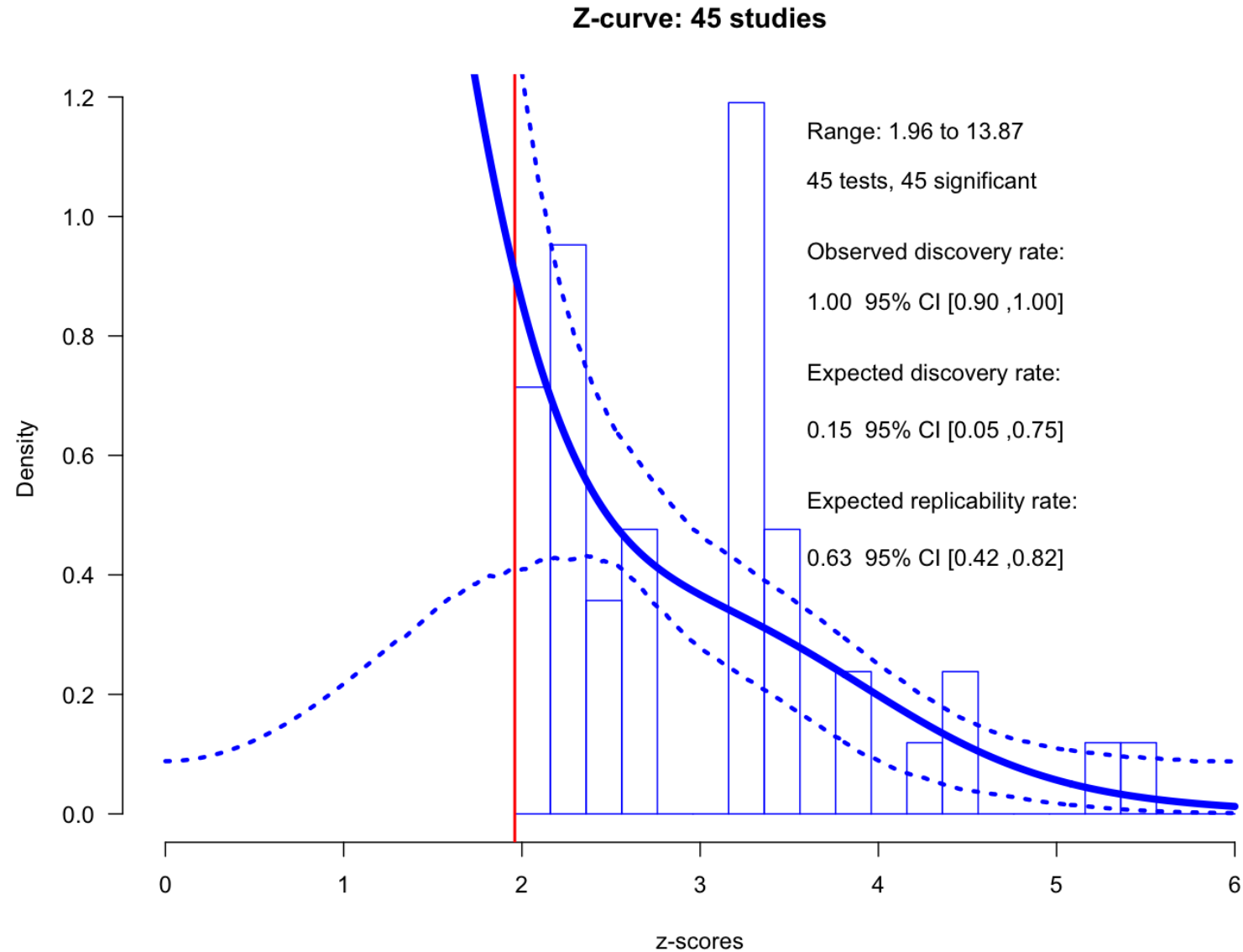
- If p-value was missing but test statistic and df were available, p was calculated from test information
- If only a threshold was reported (e.g., $p < .05$) with no other information, p was treated as the boundary value (e.g., $p = .05$)

Double-coding reliability checks

- Each paper is assigned to coder pairs by the project leader; each coder extracted data independently before comparing entries
- Initial agreement: **84%** (38/45 tests coded identically)
 - Ambiguity over which sub-comparison to extract when studies reported multiple tests
 - **Resolution:** Team voting during group meetings
 - Studies reporting both standardized and unstandardized coefficients
 - **Resolution:** Extract standardized coefficients when available; use unstandardized when no standardized estimate was reported or when it was insufficient to calculate a p-value

Results

Z-Curve Model



Z-Curve Analysis Results

k = 45 tests

ODR

OBSERVED DISCOVERY

1.00

95% CI [.90, 1.00]

All tests significant

EDR

EXPECTED DISCOVERY

.16

95% CI [.05, .73]

16% expected given power

ERR

EXPECTED REPLICABILITY

.63

95% CI [.42, .81]

63% would replicate

EFDR

FALSE DISCOVERY (SORIĆ)

.29

95% CI [.02, .99]

Up to 29% false positives

Large ODR–EDR gap signals publication bias; real effects likely present, but up to 29% of significant findings may be false positives..

⚠ small sample (k = 45) and wide EDR confidence interval [.05–.75]

Discussion

Implications for Evidential Value and Replicability

- Wide range of z-scores (1.96–11.06) → evidence of real underlying effect
- Many studies far exceed significance threshold → not pure noise
- Expected replicability rate = **0.56** (moderate)
- Strong contrast with observed discovery rate = **1.00**
- Suggests:
 - Real effect exists
 - But robustness is overstated
- Likely influenced by selective reporting or publication bias

Sensitivity to Inclusion Criteria and Analytic Assumptions

- Based only on significant results (ODR = 1.00)
 - may inflate discrepancy with expected discovery rate
- Mixed statistical tests (ANOVA, t , χ^2 , LME, r)
 - z-conversion requires assumptions
- Assumes independence across studies
 - may be violated (shared paradigms / labs)
- Small sample ($k = 45$)
 - sensitive to extreme values (e.g., $z > 10$)

Limitations of Z-curve and Distribution-Based Methods

- Uses only significant results
→ does not capture full evidence base
- Depends on assumptions about effect size distribution
→ potential model misspecification
- Detects bias but cannot identify its source
→ publication bias vs p -hacking unclear
- Sensitive to small sample size ($k = 45$)
→ influenced by extreme z -values
- Focuses on significance, not effect size
→ limited insight into practical importance

References

- Kousta, S. T., Vigliocco, G., Vinson, D. P., Andrews, M., & Del Campo, E. (2011). The representation of abstract words: why emotion matters. *Journal of Experimental Psychology. General*, 140(1), 14–34. <https://doi.org/10.1037/a0021446>
- Kroll, J. F., & Merves, J. S. (1986). Lexical access for concrete and abstract words. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 12(1), 92–107. <https://doi.org/10.1037/0278-7393.12.1.92>
- Paivio, A., & Begg, I. (1971). Imagery and associative overlap in short-term memory. *Journal of Experimental Psychology*, 89(1), 40–45. <https://doi.org/10.1037/h0031193>
- Wickens, D. D., & Engle, R. W. (1970). Imagery and abstractness in short-term memory. *Journal of Experimental Psychology*, 84(2), 268–272. <https://doi.org/10.1037/h0029108>

Thank You!

Q & A